

LAB # 10**IMPLEMENTATION OF ARRAYS****Lab Tasks**

- 1- Write a program to print elements of an array in reverse order.
- 2- Write a program that accepts temperature of 7 days from user and print there average using array
- 3- Write a C++ program to add two matrix by using multi-dimensional arrays.
- 4- Write a C++ program to find transpose of a matrix by using multi-dimensional arrays.
- 5- Use two dimensional arrays to store given monthly sale chart of different employees during 1st quarter of year 2021.

	Jan	Feb	Mar	Apr
Emp1	3,20,000	5,56,000	4,98,000	4,79,000
Emp2	5,25,000	4,56,000	5,50,000	4,79,000
Emp3	4,30,000	3,90,000	3,75,000	3,20,000
Emp4	3,25,000	4,59,000	4,55,000	4,95,000
Emp5	4,80,000	5,00,000	4,35,000	4,40,000
Emp6	5,90,000	5,70,000	5,20,000	4,25,000

Now, give function definition only that;

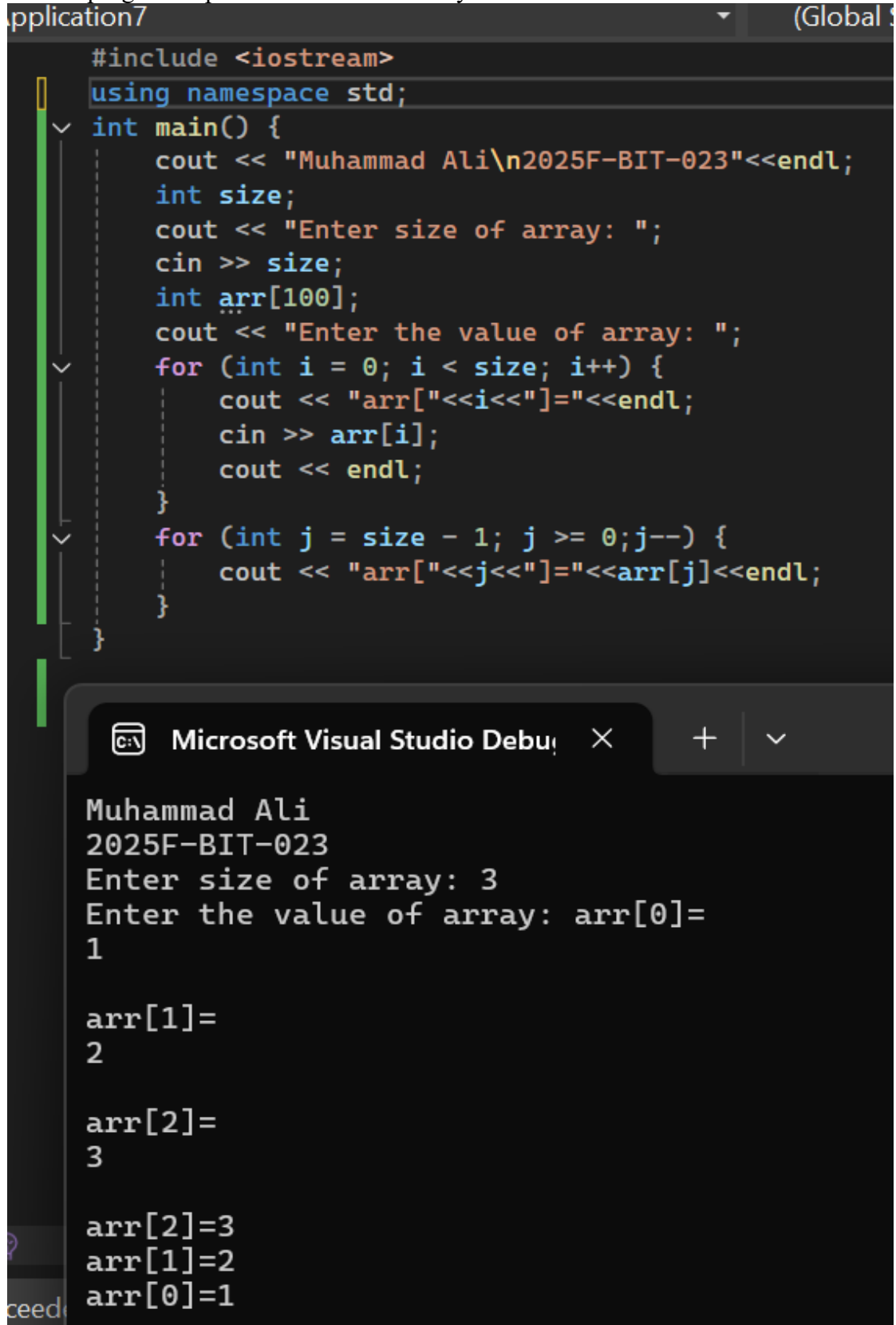
- (i) Searches the best employee with highest overall sale during first quarter, so that a letter of appreciation can be awarded to him.

void HighestSale (long [] []);

- (ii) Takes month number as input, and gives employee number with lowest sale, so that a warning letter can be issued to him to improve his performance.

void PoorPerformance(long [] []);

- 1- Write a program to print elements of an array in reverse order.



The image shows a screenshot of a C++ program in Visual Studio. The code is in a file named 'application7.cpp'. It includes the <iostream> header and uses the std namespace. The main function prompts the user to enter the size of an array (3) and then enters the values of the array (1, 2, 3). The program then prints the array elements in reverse order (3, 2, 1).

```
#include <iostream>
using namespace std;
int main() {
    cout << "Muhammad Ali\n2025F-BIT-023"<<endl;
    int size;
    cout << "Enter size of array: ";
    cin >> size;
    int arr[100];
    cout << "Enter the value of array: ";
    for (int i = 0; i < size; i++) {
        cout << "arr["<<i<<"]="<<endl;
        cin >> arr[i];
        cout << endl;
    }
    for (int j = size - 1; j >= 0; j--) {
        cout << "arr["<<j<<"]="<<arr[j]<<endl;
    }
}
```

Microsoft Visual Studio Debug Console Output:

```
Muhammad Ali
2025F-BIT-023
Enter size of array: 3
Enter the value of array: arr[0]=
1

arr[1]=
2

arr[2]=
3

arr[2]=3
arr[1]=2
arr[0]=1
```

- 2- Write a program that accepts temperature of 7 days from user and print there average using array

```
66
67     #include <iostream>
68     using namespace std;
69
70     int main() {
71         cout << "Muhammad Ali\n2025F-BIT-023" << endl;
72
73         int sum = 0;
74         int arr[7];
75         cout << "Enter temperature of 7 days" << endl;
76         for (int i = 0; i < 7; i++) {
77             cout << "Temperature of Day " << i+1 << ": ";
78             cin >> arr[i];
79             sum = sum + arr[i];
80         }
81
82         double avg_temp = sum / 7;
83         cout << "the average temperature of 7 days: " << avg_temp;
84     }
85
86
87
88
89
90
```

Microsoft Visual Studio Debug Console

Muhammad Ali
2025F-BIT-023
Enter temperature of 7 days
Temperature of Day 1: 1
Temperature of Day 2: 2
Temperature of Day 3: 3
Temperature of Day 4: 4
Temperature of Day 5: 5
Temperature of Day 6: 6
Temperature of Day 7: 7
the average temperature of 7 days: 4

- 3- Write a C++ program to add two matrix by using multi-dimensional arrays.

```
Application7 (Global Scope)

#include <iostream>
using namespace std;

int main() {
    cout << "Muhammad Ali\n2025F-BIT-023" << endl;
    int a[3][3];
    for (int i = 0; i < 3; i++) {
        for (int j = 0; j < 3; j++) {
            cout << "enter Matrix A [" << i << "]" << "[" << j << "]: ";
            cin >> a[i][j];
        }
    }
    cout << endl;
    int b[3][3];
    for (int i = 0; i < 3; i++) {
        for (int j = 0; j < 3; j++) {
            cout << "enter Matrix B [" << i << "]" << "[" << j << "]: ";
            cin >> b[i][j];
        }
    }
    cout << endl << "The sum of Matrix A & B : " << endl;

    int c[3][3];
    for (int i = 0; i < 3; i++) {
        for (int j = 0; j < 3; j++) {
            c[i][j] = a[i][j] + b[i][j];
            cout << "Matrix c[" << i << "]" << "[" << j << "]= " << c[i][j] << endl;
        }
    }
}
```

```
Microsoft Visual Studio Debug Console
Muhammad Ali
2025F-BIT-023
enter Matrix A [0][0]: 1
enter Matrix A [0][1]: 2
enter Matrix A [0][2]: 3
enter Matrix A [1][0]: 4
enter Matrix A [1][1]: 5
enter Matrix A [1][2]: 6
enter Matrix A [2][0]: 7
enter Matrix A [2][1]: 8
enter Matrix A [2][2]: 9

enter Matrix B [0][0]: 9
enter Matrix B [0][1]: 8
enter Matrix B [0][2]: 7
enter Matrix B [1][0]: 6
enter Matrix B [1][1]: 5
enter Matrix B [1][2]: 4
enter Matrix B [2][0]: 3
enter Matrix B [2][1]: 2
enter Matrix B [2][2]: 1

The sum of Matrix A & B :
Matrix c[0] [0]= 10
Matrix c[0] [1]= 10
Matrix c[0] [2]= 10
Matrix c[1] [0]= 10
Matrix c[1] [1]= 10
Matrix c[1] [2]= 10
Matrix c[2] [0]= 10
Matrix c[2] [1]= 10
Matrix c[2] [2]= 10
```

- 4- Write a C++ program to find transpose of a matrix by using multi-dimensional arrays.

```

//7773
#include <iostream>
using namespace std;

int main() {
    cout << "Muhammad Ali\n2025F-BIT-023" << endl;

    int a[3][3];
    for (int i = 0; i < 3; i++) {
        for (int j = 0; j < 3; j++) {
            cout << "enter Matrix A [" << i << "]" << "[" << j << "]: ";
            cin >> a[i][j];
        }
    }

    int b[3][3];
    for (int i = 0; i < 3; i++) {
        for (int j = 0; j < 3; j++) {
            b[j][i] = a[i][j];
        }
    }

    cout << "Transpose of Matrix A: \n";
    for (int i = 0; i < 3; i++) {
        for (int j = 0; j < 3; j++) {
            cout << "Matrix[" << i << "]" << "[" << j << "]: " << b[i][j] << endl;
        }
    }
}

```

```

Muhammad Ali
2025F-BIT-023
enter Matrix A [0][0]: 1
enter Matrix A [0][1]: 2
enter Matrix A [0][2]: 3
enter Matrix A [1][0]: 1
enter Matrix A [1][1]: 2
enter Matrix A [1][2]: 3
enter Matrix A [2][0]: 1
enter Matrix A [2][1]: 2
enter Matrix A [2][2]: 3
Transpose of Matrix A:
Matrix[0][0]: 1
Matrix[0][1]: 1
Matrix[0][2]: 1
Matrix[1][0]: 2
Matrix[1][1]: 2
Matrix[1][2]: 2
Matrix[2][0]: 3
Matrix[2][1]: 3
Matrix[2][2]: 3

```

- 5- Use two dimensional arrays to store given monthly sale chart of different employees during 1st quarter of year 2021.

	Jan	Feb	Mar	Apr
Emp1	3,20,000	5,56,000	4,98,000	4,79,000
Emp2	5,25,000	4,56,000	5,50,000	4,79,000
Emp3	4,30,000	3,90,000	3,75,000	3,20,000
Emp4	3,25,000	4,59,000	4,55,000	4,95,000
Emp5	4,80,000	5,00,000	4,35,000	4,40,000
Emp6	5,90,000	5,70,000	5,20,000	4,25,000

Now, give function definition only that;

- a. Searches the best employee with highest overall sale during first quarter, so that a letter of appreciation can be awarded to him.

void HighestSale (long [] []);

- b. Takes month number as input, and gives employee number with lowest sale, so that a warning letter can be issued to him to improve his performance.

void PoorPerformance(long [] []);

```

#include <iostream>
using namespace std;

void HighestSale(long sales[6][4]) {
    int max = 0;
    int best = 0;

    for (int i = 0; i < 6; i++) {
        int total = 0;
        for (int j = 0; j < 4; j++) {
            total += sales[i][j];
        }

        if (total > max) {
            max = total;
            best = i;
        }
    }

    cout << "Best Employee: " << best + 1 << endl;
    cout << "Total Sale: " << max << endl << endl;
}

void PoorPerformance(long sales[6][4]) {
    int month;
    cout << "Enter month number (1 for January, 2 for February, 3 for March, 4 for April): ";
    cin >> month;
    month--;

    int min = sales[0][month];
    int low = 0;

    for (int i = 1; i < 6; i++) {
        if (sales[i][month] < min) {
            min = sales[i][month];
            low = i;
        }
    }

    cout << "\nEmployee with lowest sale: " << low + 1 << endl;
    cout << "Lowest Sale: " << min << endl;
}

int main() {
    cout << "Muhammad Ali\n2025F-BIT-023\n\n";

    long sales[6][4] = {
        {320000, 556000, 498000, 479000},
        {525000, 456000, 550000, 479000},
        {430000, 390000, 375000, 320000},
        {325000, 459000, 455000, 495000},
        {480000, 500000, 435000, 440000},
        {590000, 570000, 520000, 425000}
    };

    HighestSale(sales);
    PoorPerformance(sales);

    return 0;
}

```



```
Muhammad Ali  
2025F-BIT-023
```

```
Best Employee: 6  
Total Sale: 2105000
```

```
Enter month number (1 for January, 2 for February, 3 for March, 4 for April): 3
```

```
Employee with lowest sale: 3  
Lowest Sale: 375000
```